

CHAPTER FIVE

Conclusion and Future Work

5.1 Summary of Findings and Contributions

This project set out to design, train, validate, and deploy a Smart Diabetes Monitoring and Prediction System that satisfies a binding clinical minimum Recall of 0.90, operates entirely on self-reported behavioral and demographic data, and is delivered in a form accessible to clinical practitioners without requiring programming expertise. The system was built on the BRFSS 2015 dataset comprising 253,680 records and 21 features, with a binary target variable encoding diabetic or prediabetic status. A complete machine learning pipeline was implemented across eighteen sequential steps, organized into four progressive experimental phases spanning baseline training, feature engineering, feature selection, and hyperparameter optimization, followed by systematic threshold optimization and deployment packaging.

Seven classification algorithms were evaluated across three feature configurations: the original 21 BRFSS features, a domain-engineered 45-feature space, and a reduced top-15 feature subset identified through combined Random Forest MDI and Mutual Information ranking. Model selection across all 21 model-phase combinations was governed by a clinically weighted composite scoring function assigning 40 percent weight to Recall, reflecting the medical asymmetry between false negative and false positive errors in diabetes screening. LightGBM from Phase 2 achieved the globally highest composite score of 0.6944 and was selected for hyperparameter tuning.

Systematic hyperparameter optimization using RandomizedSearchCV with Recall as the optimization criterion across an 11-parameter search space identified an optimal configuration achieving a mean cross-validated Recall of 0.9095 across five stratified folds. Threshold optimization across 90 candidate thresholds identified 0.53 as the clinical deployment threshold, defined as the highest value satisfying the binding Recall constraint of 0.90. At this threshold, the final deployed model achieves a Recall of 0.9064, correctly identifying 6,362 out of 7,019 diabetic cases in the held-out test set, with a ROC-AUC of 0.8186, PR-AUC of 0.4447, Balanced Accuracy of 0.7228, MCC of 0.3114, and F1-Score of 0.4066 [9], [15], [33].

Six original contributions were made by this work,

First, a fully reproducible end-to-end machine learning pipeline for binary diabetes prediction on the BRFSS 2015 dataset was presented, representing one of the most methodologically comprehensive treatments of this classification task in the graduation project literature [7], [12], [13].

Second, a domain-informed feature engineering framework was introduced that expands 21 original survey features to 45 engineered attributes through WHO-standard BMI category encoding, composite cardiovascular and metabolic risk scores, pairwise interaction terms, polynomial expansions, and socioeconomic proxy variables, producing measurable discriminative improvements across all seven evaluated algorithms [37], [17].

Third, a clinically aligned model selection and evaluation methodology was developed that replaces accuracy-centric assessment with a medically weighted composite scoring function explicitly prioritizing Recall, providing a principled framework for model comparison in imbalanced medical classification tasks [9], [44], [34].

Fourth, a systematic threshold optimization procedure was implemented and validated, demonstrating that a deployment threshold of 0.53 achieves a clinical Recall of 0.9064, directly operationalizing the medical priority of sensitivity over specificity in population screening [33], [30].

Fifth, an empirical comparative study across seven algorithms and three feature configurations provides strong evidence that LightGBM with `scale_pos_weight` tuning consistently outperforms all competing algorithms on clinically appropriate metrics, contributing to the evidence base supporting gradient boosting for structured healthcare tabular data [15], [16], [28], [29].

Sixth, a deployment-ready pipeline bundle was packaged and integrated into an interactive Streamlit web application, demonstrating how research-grade models can be transitioned into accessible clinical tools [4], [5].

5.2 Limitations

Five limitations apply to this study and must be considered when interpreting and extending its findings.

1. The first limitation concerns the self-reported nature of all input features. Every variable in the BRFSS dataset is derived from telephone survey responses, which introduces recall inaccuracy, social desirability effects, and systematic underreporting. Respondents may inaccurately recall behavioral patterns such as physical activity frequency or dietary habits, and they may underreport stigmatized behaviors such as alcohol consumption. These biases distort the true statistical relationships between features and the diabetes outcome in ways that cannot be fully quantified or corrected post-hoc [49].
2. The second limitation is the cross-sectional design of the BRFSS survey. Each respondent contributes a single observation at a single point in time, with no longitudinal follow-up. This design precludes any analysis of individual risk trajectories over time, prevents assessment of how risk accumulates or diminishes as behavioral patterns change, and makes it impossible to draw causal inferences about the directional relationships between measured variables and diabetes onset [12].
3. The third limitation is residual class imbalance. Despite the application of three complementary imbalance mitigation strategies, the 6.18 to 1 majority-to-minority ratio in the training data means that the model's probability estimates for the diabetic class carry greater uncertainty than those for the non-diabetic class. This uncertainty is not fully captured by any single scalar performance metric and means that the model's behavior at edge-case probability values near the deployment threshold may be less reliable than at values farther from it [9], [10].
4. The fourth limitation concerns the absence of directly measured biomarkers. The entire feature set is derived from behavioral and demographic survey variables. No fasting glucose, HbA1c, lipid panel, insulin sensitivity, or other directly measured metabolic values are available. These biomarkers remain the clinical gold standard for diabetes risk assessment, and their absence places a ceiling on the predictive signal available to any model trained exclusively on survey data [3].
5. The fifth limitation is the absence of external validation. All reported performance metrics are derived from a held-out test set drawn from the same BRFSS 2015 survey population used for training. The model has not been evaluated on an independent dataset from a

different geographic region, survey year, or demographic composition. The generalizability of the results to populations outside the United States or to survey years substantially different from 2015 therefore remains unestablished [34].

5.3 Future Work and Recommendations

Six directions for future work are identified based on the findings and limitations of this study, ordered from highest to lowest expected impact.

5.3.1 External Validation on Independent Datasets

The most immediate and clinically necessary extension of this work is external validation on an independent dataset from a different geographic context or survey year. The BRFSS dataset is collected exclusively from United States adults, and the model's behavioral risk factor weights may not transfer directly to populations with substantially different dietary patterns, healthcare access profiles, or ethnic compositions. Evaluating the trained pipeline on BRFSS data from a different year, on a similar survey from a different country, or on a clinical electronic health record dataset would establish the true generalizability of the reported performance [34], [7].

5.3.2 Integration of Biomarker Features

The current system operates entirely on self-reported behavioral and demographic data. Integrating even a small number of directly measured biomarkers, such as fasting glucose, HbA1c, or blood pressure readings, would substantially increase the model's access to pathophysiologically proximal predictors. A hybrid survey-biomarker feature set would preserve the accessibility advantage of the behavioral approach while closing the predictive gap attributable to the absence of direct metabolic measurements [3], [6].

5.3.3 Advanced Imbalance Handling with ADASYN

The current pipeline uses algorithm-level cost-sensitive weighting as its primary imbalance handling mechanism. Future iterations could evaluate Adaptive Synthetic Sampling (ADASYN) as an alternative or complementary data-level augmentation strategy. Unlike SMOTE, which generates synthetic minority instances uniformly across the minority class distribution, ADASYN concentrates synthetic generation in the decision boundary regions

where the classifier finds the minority class most difficult, potentially producing more informative training augmentation for the diabetic class at the class boundary [22], [9].

5.3.4 Bayesian Hyperparameter Optimization

The current hyperparameter optimization uses `RandomizedSearchCV`, which samples configurations uniformly at random from the defined search space without learning from previous evaluation results. Replacing this with a Bayesian optimization framework such as `Optuna` or `Hyperopt` would allow the search to concentrate evaluations in regions of the parameter space that have previously yielded strong cross-validated performance, producing better configurations with the same or fewer function evaluations [42]. This is particularly relevant for the `scale_pos_weight` parameter, where the optimal value may vary significantly depending on the feature configuration and the Recall-precision operating point.

5.3.5 Longitudinal Data Integration

The cross-sectional nature of the BRFSS dataset is one of the most consequential limitations of the current work. Access to longitudinal health survey data, in which the same individuals are observed at multiple time points, would enable time-series feature engineering capturing the trajectory of behavioral risk factor accumulation rather than its instantaneous value at a single observation point. Models trained on longitudinal data could potentially identify accelerating risk trajectories years before any single cross-sectional observation would flag the individual as high-risk [12], [6].

5.3.6 Formal Clinical Evaluation and Regulatory Pathway

Translating a research-grade machine learning model into clinical care requires careful attention to deployment infrastructure, clinician trust, and the alignment of model objectives with the practical constraints of clinical workflow [48].

Future work should include a prospective pilot evaluation in a primary care or community health setting, measuring the system's performance on real patient populations, assessing clinician trust and adoption, and identifying interface design improvements that increase usability in time-constrained clinical workflows. A formal regulatory pathway assessment would also be necessary before any deployment in a regulated healthcare environment [5], [34].

5.4 Chapter Summary

This chapter concluded the Smart Diabetes Monitoring and Prediction System by summarizing the principal findings and contributions, assessing the study's limitations honestly, and identifying six prioritized directions for future work. The system successfully satisfies its primary clinical objective, achieving a Recall of 0.9064 at deployment threshold 0.53 and correctly identifying 6,362 out of 7,019 diabetic cases in the held-out test set, with a ROC-AUC of 0.8186, PR-AUC of 0.4447, Balanced Accuracy of 0.7228, and MCC of 0.3114. The five limitations identified, covering self-reported data bias, cross-sectional design, residual class imbalance, absence of biomarkers, and lack of external validation, define a clear research agenda for subsequent iterations. The six future work directions, spanning external validation, biomarker integration, advanced imbalance handling, Bayesian optimization, longitudinal data integration, and formal clinical evaluation, collectively describe the pathway from the current academic demonstration toward a clinically validated and deployable early detection tool [9], [5], [34].